BEHAVIORAL scientist

SCIENCE

Special Collection: Examining the Intersection of Behavioral Science and Advocacy

By **Cintia Hinojosa** and **Evan Nesterak**

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Over the past year, everyone's lives have been touched by issues that intersect science and advocacy—the pandemic, climate change, police violence, voting, protests, the list goes on.

These issues compel us, as a society and individuals, toward understanding. We collect new data, design experiments, test our theories. They also inspire us to examine our personal beliefs and values, our roles and responsibilities as individuals within society.

Perhaps no one feels these forces more than social and behavioral scientists. As members of fields dedicated to the study of social and behavioral phenomena, they are in the unique position of understanding these issues from a scientific perspective, while also navigating their inevitable personal impact. This dynamic brings up questions about the role of scientists in a changing world. To what extent should they engage in advocacy or activism on social and political issues?

Should they be impartial investigators, active advocates, something in between?

> ### Should scientists be impartial investigators, active advocates, something in between?

It also raises other questions, like does taking a public stance on an issue affect scientific integrity? How should scientists interact with those setting policies? What happens when the lines between an evidence-based stance and a political position become blurred? What should scientists do when science itself becomes a partisan issue?

To learn more about how social and behavioral scientists are navigating this terrain, we put out a call inviting them to share their ideas, observations, personal reflections, and the questions they're grappling with. We gave them 100-250 words to share what was on their mind. Not easy for such a complex and consequential topic.

The responses, collected and curated below, revealed a number of themes, which we've organized into two parts. Part 1 features half of the total responses and showcases those that revealed how scientists are grappling with science and advocacy broadly. This includes what it means to do good science, neutrality as critical—or an illusion, the role of identity and experience, concerns about politics and making an impact, and more.

Part 2 features the responses that explore more specific aspects of the issue and is divided into three sections. The first focuses how behavioral science and scientists relate to larger social systems and structures, like democracy, financial systems, and the people and communities they research. The second houses responses that cover the ways behavioral science might be positioned to inform advocacy, while the third contains specific ideas about the ways science and advocacy are currently coming into contact in practice.

Investigating the relationship between science and advocacy means attending to an ever-growing network of complexity and nuance. It's a task that can be overwhelming to resolve independently, yet intimidating to discuss publicly.

The goal of the collection is to inspire reflection and conversation. To bring the often hidden conversations and personal, internal dialogues into the open, hopefully helping build a shared, public understanding of the issue. Investigating the relationship between science and advocacy means attending to an ever-growing network of complexity and nuance. It's a task that can be overwhelming to resolve independently, yet intimidating to discuss publicly—especially where one might be vulnerable to passion-charged criticisms. We want to thank the contributors who took the bold step to share their thoughts.

It's important to note that we do not intend this collection to be the final word on the topic, nor do we want to prescribe any particular direction to follow. We only hope that you will approach the collection openly.

It's in this spirit that we invite you to this collection.

— Cintia Hinojosa and Evan Nesterak

Table of Contents

Note: We sectioned Part 1 in three sections of seven responses to help with navigation. We didn't categorize responses, as we didn't want to be overly prescriptive so that readers could make their own connections.

Part 1

Grappling with science and advocacy

How scientists are thinking about the intersection of science and advocacy broadly, including what it means to do good science, neutrality as critical—or an illusion, the role of identity and experience, concerns about politics and making an impact, and more.

Section 1.1

Social scientists ought to bring their knowledge to bear on social issues, but advocacy opens up a danger that is parallel to scientists who have a financial interest in a drug whose efficacy they are testing: the temptation to spin, fudge, data-snoop, p-hack, or outright fabricate their results to ratify their political positions.

We have every reason to believe that the motive to glorify oneself and one's political tribe is as powerful as pecuniary self-interest—if social scientists were primarily motivated by money, they would not have chosen academia in the first place—so the conflicts of interest are just as powerful. Probably more so, since everyone knows that money corrupts, but few people acknowledge that their political and policy opinions may be creeds of their tribe rather than obviously correct truths. As with drug testing, we ought to be suspicious of political partisans recommending policies unless their reasoning is subjected to the strictest scrutiny, alert to the potential for bias, and evaluated by fair, politically neutral criteria.

***Steven Pinker** is a professor in the department of psychology at Harvard University and author of books on language, cognition, and human nature,* *including* Enlightenment Now.

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Researchers are often afraid of being involved in advocacy because they fear losing the perception of neutrality. But this perception is an illusion. Researchers are people too and bring their own values and norms to their work. When we choose which hypothesis to test, which counterfactuals to estimate, what statistical tests to use, and which studies are worth publishing, we are making decisions that reflect who we are as people. The key is to be transparent about our biases, as opposed to hiding them under a cloak of assumed neutrality. Given this, the question becomes whether researchers are interested in using their knowledge to improve the world we live in. Few would argue against the need for evidence-based policy, and a natural corollary is that those who best know the evidence are an active part in determining what policies are enacted.

***Diogo Veríssimo** is a research fellow at the University of Oxford and the director of conservation marketing at On the Edge Conservation.*

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Researchers often take positivist approaches to questions. That is, they try to explore and understand "what is." This may then inform normative perspectives, or "what should be," which may then naturally translate into changes in policy, law, or practice. However, if ex ante, there is only one potentially interesting or "correct" result to a question, then it was not a true research question to begin with and could be reasonably construed as an exercise solely to confirm one's own biases and ideologies. We want to be expanding our understanding of the world and not just setting out to reaffirm our priors.

***Anjali Adukia** is an assistant professor at the University of Chicago Harris School of Public Policy.*

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On balance, I think that scholars getting involved in advocacy is generally a mistake, at least for the area in which they are doing research. This is not to say that scholars can't try to keep policy makers informed of research data, though they should strive to do so in the most neutral and objective ways without pushing for specific policy outcomes.

The inclination to become scholar-advocates is very human and understandable. We want to help, and we think the data we produce can be helpful for making policy decisions. But it also opens us up to sunk costs—we become invested in particular outcomes and, in doing so, too easily end up with skin in the game. Much of psychology's replication crisis boiled down to researcher expectancy effects and p-hacking, and that can only get worse once scholars get morally invested in certain outcomes.

I saw this happen in my own field where some scholars, and even the American Psychological Association, misinformed the public about the data on violence in video games. We need to learn from that example and keep as "pure" as we can as representatives of the data, not advocates for specific policies.

Christopher J. Ferguson *is a professor of psychology at Stetson University.*

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Behavioral scientists are trained to look for, make sense of, and reduce bias in the decision making of others. We also try to eliminate bias in our approach to crafting research questions that are linked to complex social issues. Our individual identities and values play a role in how we do our work.

As a researcher who engages at the intersection of social policy and psychology, I have typically tried to remain "neutral." I have come to terms with the reality that this, in and of itself, reinforces the structures that I actually want to dismantle and re-envision. Now, I am working to unapologetically acknowledge and draw on my own lived experience as a Black woman in my work. My identity is central to how I shape my research and engage in the classroom.

It is entirely possible to engage in thorough and careful evidence-building in one's scholarship, and to also find spaces to be an advocate for what is just. The mechanisms of peer review in our scientific processes already provide a critically important system of checks and balances. The emerging culture of open science creates excellent new mechanisms to guarantee the rigor of evidence.

Behavioral scientists should embrace these tools that improve our scientific methods and culture, while also speaking up in the spaces that call for our expertise. The challenges that our world faces are too large and too urgent for us to behave otherwise.

Crystal C. Hall *is an associate professor of public policy and adjunct associate professor of psychology at the University of Washington.*

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Science and advocacy don't mix. Scientists should embrace a mantra first formulated by Stanford's Paul Saffo in the context of forecasting: "strong opinions, weakly held." In science, "strong opinions" refers to the essential requirement to back any claim with solid evidence; "weakly held" means that, when this evidence ceases to support the claim, or alternative evidence is found to be superior, one should change one's mind and stop making the claim.

A good activist will have strong opinions, backed by evidence. But a good activist, inevitably, is also deeply invested in the cause she or he supports. This goes together with a powerful sense of ownership, a profound conviction verging on the religious. The issue with this attitude lies with the second half of the mantra. A strong sense of ownership is associated with several potential sources of cognitive distortion, from the sunk cost fallacy and confirmation bias to the endowment effect and motivated reasoning. Thus, the intense belief in a cause that characterizes the activist makes it very hard for an activist scientist to hold her or his strong opinion weakly, to consider contradicting evidence dispassionately, and to be ready and willing, without any resistance, to ditch the cause. And that is anathema to science.

That said, there is one form of activism that I would endorse for scientists: activism that promotes and champions the scientific method.

***Koen Smets** is an organization development consultant, senior adviser to the BVA Nudge Unit, and an adjunct assistant professor at Saint Louis University, where he teaches a course in ethical and evidence-based decision-making.*

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In a recent survey on the perspectives and civic engagement of members of the Union of Concerned Scientists (UCS), the Science Advocacy Working group found that Multiethnic and non-White scientists were more likely than their White counterparts to engage in science advocacy. They were substantially more likely to post to social media, work on projects with a professional association or community group, and provide expertise for an advocacy group. While White scientists were more likely to donate to a nonprofit or contact an elected representative, non-White scientists were more inclined to show political participation through direct community engagement at a public meeting, participation in a protest/march/vigil, or involvement with an advocacy-aligned project through their employer.

Given that UCS membership skews towards those who are more politically inclined, these distinct engagement styles between White and non-White scientists highlight important differences. The present-day elite- and male-dominated standards of impartiality and disinterestedness associated with traditional ways of maintaining credibility within the profession are not necessarily the narrow values that many scientists of color operate under. This desire to engage in advocacy within their profession may be in direct alignment with a deeper understanding, by virtue of lived experience, of the role that power, politics, science, and trust plays in marginalized communities. Their engagement work reflects the need to have a more integrated approach to science and advocacy, in which part of "doing" science is intentionally connecting, communicating, and translating research to impact policy.

***Meghan Lynch** is pursuing a Master's in public policy at the University of Maryland, Baltimore County and is the graduate assistant for the Union of Concerned Scientists Strengthening Science Advocacy Working Group. **Barbara Allen** is a professor in the Department of Science, Technology, and Society at Virginia Tech's Washington, D.C. area location and is the author of* Uneasy Alchemy: Citizens and Experts in Louisiana's Chemical Corridor Disputes*. **Dana Williamson** is an environmental health fellow hosted at the U.S. Environmental Protection Agency, leading the program evaluation efforts of the Scientific Integrity Program. She is also an Agent of Change Fellow in affiliation with the George Washington University Milken School of Public Health.*

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Section 1.2

Scientists can be advocates in the policy arena, but they will have a greater impact on public policy if they are not. In fact, we need more scientists to take a stand for a non-advocacy approach in the policy arena. The reputation of science (and scientists) will improve and we'll ultimately get better public policies.

For more than 10 years, I have worked at the intersection of research and policy at a major university. I have met with hundreds of legislators and other policymakers with the goal of helping them appreciate and use research in their work. I have helped more than 150 faculty members from my campus and other universities communicate their research in a way that will be trusted and used by policymakers.

What I have learned is that elected officials will first look for a researcher's underlying biases. They will look at who funded your research, whether you have written op-eds naming and shaming individuals or groups, whether you appear to have an agenda, and what you post on social media. Researchers are often unaware of how easily they slip into advocate mode and how much care must be taken to appear unbiased and objective. Yet being known as an honest broker and a non-advocate is essential for long-term success in the policy world.

The most successful researchers—those trusted on both sides of the aisle—present the literature fairly and dispassionately and lay out the consequences of various policy options. They then stop and listen.

Heidi Normandin *is the director of legislative outreach at the University of Wisconsin–Madison La Follette School of Public Affairs.*

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As an African-American doing research on police-community relationships, I have had to develop a sophisticated understanding of how to advance research on topics that blur the lines between my personal and professional life, and I feel a responsibility to help others do the same. In my work, I have continued to rely on the scientific method to develop practices that will yield positive outcomes for our communities. We need concrete, scientifically-backed, and reliable practices that police and their communities can use to keep themselves safe. We need to all work together on the complexity of this problem from a diverse array of perspectives. Researchers cannot avoid difficult topics out of fear of discourse or discord.

Kyle Dobson *is a postdoctoral researcher at the Texas Behavioral Science and Policy Institute at the University of Texas at Austin.*

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Although neither scientists nor policymakers are incentivized to speak to one another, their collaboration is necessary if we want high-quality, rigorous, objective research to influence policy. Many research articles with important information to change people's lives are left in the pages of academic journals. Policymakers could seek out such research, but instead mostly use the evidence that comes across their desks from advocacy groups. Without scientists advocating for their research to be heard in large audiences, even the most important and impactful findings may never make change.

We are best positioned to translate our research into actionable insights because we know both the magnitude and limits of our impact. If our advocacy is rooted within honest reflection of the conclusions we can and cannot draw, then it can be a powerful way to get high-quality, rigorous, objective research to be considered in important policy decisions. If not, that is advocacy separate from science.

Jamie M. Carroll *is the associate project director for the Texas Behavioral*

Science and Policy Institute at the University of Texas at Austin. She earned a Ph.D. in sociology.

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It might be comforting to believe that science and advocacy should be separate. But that is not how science works. Science—especially social science—does not operate in a vacuum. Scientists choose their lines of inquiry, at least, in part, based on their personal interests. The researchers at the Jameel Poverty Action Lab are looking for innovative ways to reduce poverty throughout the world. To bar them from the advocacy arena would belie their purpose.

The challenge, of course, is that scientists must construct an impenetrable wall between their objective, positive research from their subjective, normative personal perspective. Unfortunately, the quest for findings that are statistically significant and in the "right" direction can lead researchers to engage in ethically questionable activities (a dropped outlier here, a fishing expedition there). Although such behavior can occur anytime, it is especially problematic when the researcher wants to use the results for advocacy.

The remedy might be not to bar scientists from advocacy, but for universities, think tanks, and funders of research to promote "adversarial collaborations," where scientists with opposite opinions on a topic (say, climate change) design a research project with agreed-upon objectives, methods, and measures of success. The "winner" gets to promote the research results, and the "loser" gets to rethink previously held positions.

Doug Hough *is a retired associate scientist at the Bloomberg School of Public Health at Johns Hopkins University and author of the book,* Irrationality in Health Care.

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The question "To what extent should social and behavioral scientists engage in advocacy or activism on social and political issues?" is a complicated one. On the one hand, thinking about the normative meaning of the question, the answer of course should be yes. Climate scientists, who spend their lives understanding the dangers of climate change, should advocate for change. Political scientists should act to stop democratic backsliding in the United States.

These examples seem like absolute no brainers. But the reality, of course, is that these things do not happen in a vacuum. The United States is a polarized country, where once an issue becomes politicized, the public quickly polarizes along partisan lines. I have seen this in my own research on climate change attitudes and, more recently, COVID attitudes. Furthermore, Republican trust in science already is diminishing. Increased science advocacy, which would presumably take place within issue domains aligned with the Democrats, like climate or pro-democracy reforms (which, unfortunately, is an indictment of the modern GOP) could conceivably reduce the impact of science on our society, and further marginalize scientists among conservatives, who would increasingly view scientists simply as a constituency of the left.

The question we should be asking is therefore this: do the benefits of advocacy by scientists outweigh the risks of completely polarizing science as a domain? Because, once something is polarized, it is incredibly difficult to "unring the bell."

Dominik Stecula *is an assistant professor of political science at Colorado State University.*

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Political advocacy from scientists can exacerbate political polarization. But, that doesn't mean it isn't worth doing. In 2017, I published research suggesting that Americans' views toward scientists became more politically polarized, following the March for Science. Ideological conservatives came to hold more negative views toward scientists, while liberals held more positive views.

Since then, one of the questions I've grappled with the most is whether or not advocacy is "worth" the potential cost. The answer, in my view, is yes. Science is, and has always been, political. Government funding for our research, efforts to restrict what we can say in the classroom, and the extent to which our work influences evidence-based policies implies that scientific research both shapes and is shaped by politics.

Consequently, scientists may from time-to-time need to take action; to march for science, state our opposition to efforts to limit our policy influence, and (as is becoming increasingly common) to even run for political office.

Advocacy risks partisan backlash. However, I see backlash not as a reason to forego political action, but to instead be cognizant of its potential costs. Public opinion toward us and the work we do may suffer, but advancing our collective interests may well be worth the cost if it allows us to be better educators, researchers, and citizens.

When science is under attack, it's okay to fight back. We simply need to recognize that we might [get a scratch or two] in the process.

Matt Motta *is an assistant professor of political science at Oklahoma State University.*

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Behavioral science is political. For (behavioral) scientists, to be part of policymaking or not is a false dilemma. Our decisions on what to study, where to allocate time, or with whom to work, all have political consequences. The choice is then to what extent we want to be involved in the political implications of our work.

Scientific research is also only as useful as its ability to enrich social and political debate and create knowledge for the development of policies with social impact. In this context, advocacy is one way of taking a political stance based on quality scientific knowledge, although not the only one. For instance, creating a lab dedicated to the application of behavioral science to the field of diversity and inclusion is a clear political stance.

Simultaneously, policymakers are, by default, choice architects. How regulations and policies are written defines their framing and affects status quo bias. Policymakers decide which topics are priorities, where to allocate resources and, ultimately, which citizens are supported by the government and which are left behind. In the process, they are influenced by several factors, which they might be aware of or not. Behavioral science can be part of it, informing decisions, and advocacy can be the means.

However, when policies can threaten the lives and rights of minoritized groups, supporting them with evidence-based arguments should be a duty to researchers. This is the belief that led us to work in the intersection of behavioral science and diversity and inclusion.

Luana Almeida, João Matos, Caio Cruz, Carlos Mauro *are members of the Behavioral Insights for Inclusion Lab (BiiLab)/CLOO – Behavioral Insights Unit.*

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Section 1.3

The question is posed, should scientists leave policy decisions up to others? And my answer is a hearty no, for two reasons.

First, scientists are affected by policy. We're human. Spending a day in the lab or out in the field doesn't prevent us from being affected by policy changes and decisions made in offices half a continent away. Congress is responsible for allocating taxpayer money to scientific research and politicians appoint the leaders of our scientific agencies—which means policy is, to some degree, involved in deciding what research even happens.

Second, our work as scientists affects policy. We investigate problems and produce data in an attempt to answer questions. Some would argue that we have a responsibility to ensure our work is understood in good faith. If we don't put forth an effort to do that, we open ourselves (and our data) up to gross misinterpretation undertaken to achieve a political goal.

Science and policy have a complex symbiotic relationship. They inform and support each other, for better or for worse. And as long as that relationship exists, scientists should have a seat at the table and be involved in policy decisions.

Gates K. Palissery *is a Ph.D. student in the Translational Biology, Medicine, and Health (TBMH) program at Virginia Tech.*

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Whether you voted for them or not, everyone has the right to engage with their elected officials, the folks who have been elected to represent their interests. I argue that it is even more important for social and behavioral scientists to engage with lawmakers. Many scientists are taught that it is not appropriate or that it somehow undermines the science to wade into the waters of policymaking or politics. While there can be some pitfalls—such as the appearance of partisanship—the benefits greatly outweigh the risks.

As a social and behavioral scientist, you bring tremendous value to your engagement with elected officials. Not only are you a constituent (read: a VOTER); you are a constituent with expertise that can meaningfully inform policymaking. In other words, when it comes to informing policy, you have power as a constituent *and* a scientist.

Another argument for why scientists should engage in policymaking is a practical one: if you are not making a case for policy that is based on evidence, you should not expect anyone else to do it for you. It is not hard to imagine a world where policy is made with little or no input from the scientific community. Sadly, it happens all the time. The best thing scientists can do to help is simply reach out to lawmakers and offer to serve as a resource. You may be surprised how many will jump at the offer.

Wendy A. Naus *is the executive director of the Consortium of Social Science Associations (COSSA), a nonprofit advocacy organization promoting social and behavioral science research to inform federal policies.*

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Looking at the current public discourse on climate change, COVID-19 pandemic response measures, and other issues, it seems more important than ever to explore the role of science in advocacy and policy decisions. How science and advocacy mix has been studied in terms of costs to scientists' credibility (an effect that is actually fairly small), but what is the cost of not taking a stance, and what could be gained?

Science and advocacy often operate in very different ways. Scientists tend to make precise, nuanced arguments, whereas advocacy seeks to unite people behind a big, bold statement. Science operates on longer timescales, while advocacy requires quick reactions to events. While science aims to uncover an objective truth, advocacy looks to influence value-based choices.

However, advocacy, policy, and science intersect in research funding, research questions, and the application of findings. For example, in my area of work, the policy-level decision to invest in research on gender-based violence in schools has enabled better understanding of its impact, which in turn informs education sector advocacy. Scientists, activists, and policymakers have an interest and a responsibility to engage with each other in a way that acknowledges and respects each domain's needs and constraints.

Scientists' expertise is crucial in translating research into better policy decisions, and scientists can improve credibility and uptake. When scientists withdraw from public debate, policy decisions will still be made. We need scientists' participation to make sure those decisions are grounded in the best available evidence.

Katharina Anton-Erxleben is a neuroscientist working in international development with over 15 years of experience across academia, government, and the nonprofit sector.

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What makes me conduct tedious, half-a-decade-long studies, shrug off journal rejections and endure long working hours for the pay that many of my students get two to three years after graduation? The desire to make the world a better place. I work on environmental sustainability, health, and charitable giving because I believe that these are areas in which we can improve the lives of others by understanding how to design better policies that promote these goals.

I think it is desirable that researchers in any field choose questions based on what they are passionate about—how else will they stay motivated for 40 years? However, this passion shouldn't affect the chosen research method or the results. Good peer-review and editors will usually keep this problem in check. In addition, experts should dare to get more involved in the policy discussion. They are the ones who know most on their topic and are the least dependent on popular opinion—they neither need votes nor do they sell anything. If great social science research doesn't make it into the policy discussion, then what is it for?

Christina Gravert *is an assistant professor of economics at the University of Copenhagen and cofounder of Impactually, a behavioral science consultancy.*

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We are going through a climate crisis. We understand the causes of climate change and the impacts, and with this knowledge comes power to shape solutions. We need to act, activate, and advocate for action. Advocating for solutions does not discolor science, but in my view enriches it for the greater good. It is the responsibility of the scientist to communicate our findings, the uncertainties of what we know and what we do not know, and pathways to deal with the problem. Science for the sake of science alone is not enough. Connecting our work to solutions can give our efforts more meaning. We have no time to lose.

Shahzeen Attari *is an associate professor at the O'Neill School of Public and Environmental Affairs at Indiana University–Bloomington.*

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Most of the time, behavioral scientists get involved when an audience intends to act, but never quite manages to. Sometimes, the goal is straightforward enough—people want to generally be healthier and wealthier. Make 401ks opt-out; remind patients to take their medicine properly.

But sometimes it's murkier. When we ask people to take action on climate, we ask them to do things that do not necessarily make them healthier and wealthier this week or this year (or sometimes, even ever), but benefit others instead—and in vague, far-off ways.

Deciding where the intention-action gap lies, and what the "right" action is to take, can quickly bring us into advocacy territory. Do people really want to buy cars with better fuel economy, move their money into a green bank, buy offsets, eat less meat? How far do we go to nudge them into that "right" action?

If we just try to debias them instead, what if they decide that all pro-social actions are a waste of their time and resources, because ultimately every individual behavior does, indeed, only amount to an infinitesimal part of what's needed?

I've grappled with this in our Bank for Good work, where climate and racial justice advocacy play a strong and critical role in a campaign to help people switch to sustainable banks. Without taking some sort of advocacy stance, we risk getting people stuck in a whole lot of System 2 navel-gazing. What is the role of objectivity when the whole world is on fire?

Sarah Welch *is a vice president at ideas42, where she helps lead behavioral innovations in two focus areas: tackling climate change and advancing environmental sustainability, and improving how donors give to charity.* ***Erin Sherman*** *is a vice president at ideas42, where she focuses on helping people and systems mitigate and adapt to climate change.*

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Within each generation of research psychologists, soul-searching questions emerge regarding whether and how our research can make a difference in the "real world." Collectively, then, we might wonder why psychological science has not yet had the level of public impact it could conceivably have.

I believe many psychologists struggle with some fundamental ambivalence about conducting socially relevant research—vacillating between wanting their research to contribute to meaningful social change and fearing that doing so may undercut its perceived scientific integrity.

However, conducting socially relevant research and maintaining scientific rigor need not be framed as incompatible goals; rather, our fears may compel us to perceive incompatibility in these goals. In part, we may fear losing status or credibility as rigorous scientists by attending to the social relevance of our work; alternatively, we may fear losing control as we seek to understand how social processes function in more complex, applied settings, or we may fear losing control over how other people use or interpret our research findings as we "give psychology away."

As behavioral researchers, we know a lot about what happens when we fear loss: we focus more attention on potential negative outcomes and become less willing to explore and take risks. We should therefore reframe how we think about psychological research and its potential for public impact, to consider what we stand to gain from attending to its social relevance—or alternatively, what we stand to lose by not attending to the social relevance of our research.

Linda R. Tropp *is professor of social psychology and faculty associate of public policy at the University of Massachusetts Amherst. She directs the UMass Public*

Engagement Project and is the editor of Making Research Matter: A Psychologist's Guide to Public Engagement*.*

Part 2

Specific aspects of the science and advocacy issue

The responses that explore more specific aspects of the issue and is divided into three sections. The first focuses how behavioral science and scientists relate to larger social systems and structures. The second houses responses that cover the ways behavioral science may be positioned to inform advocacy, while the third contains specific ideas about the ways science and advocacy are currently coming into contact in practice.

2.1: Structures and systems where behavioral scientists operate

Behavioral scientists have a responsibility to advocate for democracy in policy making, rather than for managerial technocracy in the pursuit of policy outcomes. Policy questions have no doubt benefited tremendously from the insights behavioral research brings to bear regarding human psychology, (ir)rationality, and behavior. Yet when such scientific insights are accompanied by ideologies like "libertarian paternalism," which exalts putatively apolitical behavioral scientists into positions of unaccountable power while diminishing democratic control over policy, behavioral scientists widen the gap between the average citizen and their government. I doubt that autocratic regimes like the United Arab Emirates would establish a "Nudge Unit" if they believed it would emancipate their citizens or fail to extend the coercive power of the state.

Such situations occur when behavioral scientists naively imagine they can be apolitical technicians of public policy. When scientists with such views find their "apolitical" work coopted for malevolent ends by public and private actors, the best they can muster is to call it "sludge." That is not enough. "Politics" is merely the word we use to describe the processes by which power and resources are distributed, and as such no one interacting with government policy can possibly be apolitical. Believing so just means one's politics are the reinforcement of existing power hierarchies.

All behavioral scientists have a responsibility to ask whether their work reinforces existing hierarchies and inequalities, or whether their work expands democratic values, representation, and influence over the distribution of power and resources in society.

Jeffrey Lees *is a visiting assistant professor at Clemson University's Wilbur O. and Ann Powers College of Business and the founder of the Union of Democratic Labs.*

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Social and behavioral scientists often embed their beliefs into what is studied, how they study it, and how the evidence is interpreted. It is no secret: data is a value-laden judgement.

Take health, for example. Thousands of studies operationalized health into measures of weight loss or worse still, days at the gym. Such studies not only embed assumptions of bourgeois wealth, in both time and material excess, but also position their findings as addressing "the obesity epidemic"; an epidemic so intertwined in racist ideologies that dissemination of studies which fail to acknowledge its narratives are harmful to marginalized communities.

When considering whether behavioral and social scientists should engage in advocacy, we might do well to ask: What are we advocating for by studying *this* phenomenon *this* way? Otherwise, we risk the perpetuation of "make a difference" scientists conducting studies that uphold hegemonic, and often oppressive, cultural narratives.

In contrast, social and behavioral scientists can also foster new narratives that liberate people from oppressive narratives and cultivate collaboration between seemingly disparate groups. Behavioral science research, for example, revealed that Americans share a similar preference for an energy future moving toward renewables regardless of their political affiliation—and this is just the tip of the iceberg for how social and behavioral scientists can advocate for a society centered on joy, love, and well-being.

Social and behavioral scientists must be purposeful about the beliefs they embed into their research. It is not *if* scientists blend advocacy and science, it is interrogating *what* scientists advocate for through science.

Katelyn Stenger *is a Ph.D. fellow in the Behavioral Science for Sustainable Systems program at the Convergent Behavioral Science Initiative at the University of Virginia.*

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Behavioral science has opened its arms (and purse) to harness psychological insights in order to better understand inequities and disparities. Ironically, it is these behavioral scientists that can be the most inert when it comes to advocacy—deflecting the call to take action because the -ism that they research buys them automatic advocacy points. How many social scientists with papers on poverty volunteer with low-income households? How many male social scientists have won awards for their inspiring gender research while engaging with women's rights advocacy?

This leaves us with a troubling double-edged sword; under the surface of this well-intentioned veneer of "social science for good" is the reality that research with marginalized groups is inevitably used to further one's own career while simultaneously doing little to help the community at the center of it all. Academic pursuits surrounding a "cause," while applauded by our research community, can too often overlook the sobering reality that successful careers are built on the backs of forgotten people and communities whose experiences have shaped our very research programs. For this reason, advocacy should not be perceived as a choice, but as a core tenet of the research process.

Beyond an appeal to morals, an effort to build a culture of meaningful interactions with the communities we research is ultimately good for science. It ensures that we're asking the right questions. Science and advocacy are not at odds with each other, and nor do they just co-exist, they are two sides of the same coin.

Radhika Santhanagopalan *is a joint Ph.D. student in psychology and behavioral science at the University of Chicago.*

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When India was under British rule, the government was concerned about the number of cobras in Delhi. They introduced a behavioral intervention: citizens could claim a cash bounty for every dead snake they turned in. The scheme worked so well that it was eventually scrapped. At this point, the truth was revealed—the citizens had been breeding the snakes to claim the bounty, and the now-worthless snakes were subsequently set free, making the problem even worse.

The road to hell is paved with good intentions. For example, the use of COVID-19 fear messaging, advised by the behavioral scientists on the U.K.'s Scientific Advisory Group for Emergencies (SAGE), may well have exacerbated the burgeoning mental health crisis.

Besides which, who gets to decide on the greater good? Academia, especially psychology, is rather lacking in diversity of thought: a review co-authored by Jonathan Haidt found that liberals are increasingly outnumbering conservatives in psychology, by fourteen to one at the latest count. SAGE's behavioral scientists even count a Communist party member among their ranks. There's nothing inherently wrong with that, of course, but there does seem to be a certain political slant. At any rate, a key tenet of behavioral science is that we, all of us, are prone to irrationality. Doctors' prescribing decisions are influenced by biases like diffusion of responsibility; and a recent study found that politicians are more likely than citizens to be influenced by the sunk cost fallacy.

Ultimately, who nudges the nudgers? It would be inaccurate and dangerous for behavioral scientists to ordain themselves the grand poohbahs of reason; to assume that they are rational when everyone else is irrational, and therefore that only the government's Ministry of Choice is fit to decide people's decisions for them.

***Patrick Fagan** is a part-time lecturer in consumer psychology, author of the marketing psychology book* Hooked*, chief scientific officer at Capuchin Behavioural Science, and formerly the lead psychologist at Cambridge Analytica.*

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It is critical that scientists act as advocates for evidence-based approaches toward solving social, political, and environmental challenges—social and behavioral scientists hold special responsibility, as our most dire issues are anthropogenic. Around 70 years ago, the Great Acceleration began, a period in our species' history that saw unparalleled growth and activity that wrought such an impact on our planet that many have declared this the Anthropocene, the human epoch. The Fourth Industrial Revolution is transforming not just the way we work, but the essence of the human experience. Together these human trends have resulted in social, political, and environmental problems that not only threaten our well-being but our continued existence.

As George Miller argued in 1969, "The most urgent problems of our world today are the problems we have made for ourselves … They are human problems whose solutions will require us to change our behavior and our social institutions."

While our field's methods can be improved, the behavioral science community is committed to understanding and changing human behavior through rigorous evidence-based approaches. Non-scientific positions are less connected to the truth of these challenges and less likely to yield the right solution.

Nathaniel Barr *is a professor of creativity and creative thinking at Sheridan College and a scientific advisor at BEworks.*

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Every scientific effort is aimed to move human understanding from an initial state to a more advanced one. Change is central to this endeavor, and this implies that the scientist actively supports knowledge-based transformations of human mindsets.

Yet, new knowledge often conflicts with existing structures, which may become obsolete in light of scientific advances. Although outdated, these structures can remain socially and culturally consolidated, perpetuating the status quo and hindering human progress. Therefore, the engagement of scientists in transforming such structures is just as necessary as participating in the research itself. It is, indeed, another dimension of the advancement of knowledge. Advocacy is precisely about fostering such structural changes: it is about making space for progress within society and government.

However, scientists often overlook this essential dimension of activity. This responsibility can no longer be ignored.

David Gutierrez *is a doctoral fellow in the Convergent Behavioral Science Initiative at the University of Virginia.*

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Advocacy often informs, and indeed leads, science, particularly in relation to civil rights. For example, African American advocates were pushing for voting rights and an end to Jim Crow policies while mainstream scientists were still using junk science to justify racist policies. Only when advocacy began to change the national conversation did science follow, eventually determining the entire concept of race as socially constructed.

Science, whether or not we would like to admit it, is intimately bound to advocacy. Science needs advocacy to serve as an ethical compass of sorts. And advocacy needs science to back up its ideas with evidence in a systematic way. Scientists should embrace this relationship to become scientist-advocates, using science to illuminate facts and theories, while using advocacy to direct where science should focus its attention to promote ethics and equity. Otherwise, those advocating for harmful policies will use "science" to justify those policies, contributing to injustices like the restriction of voting rights.

Judson Bonick *is a senior behavioral researcher on the Health team at the Center for Advanced Hindsight at Duke University.*

Whether science and advocacy can be good bedfellows will likely depend on what advocacy movements we're talking about. As a DEI consultant, I work with multiple organizations that have non-partisanship as one of their core tenets, such as public media, federal contractors, or research institutes. Part of my client work entails making sure organizational leadership has a complete understanding of the three Ds: diversity, dehumanization, and democracy.

Diversity means people get to be different, including in their political leanings. However, the line to draw in the sand is against dehumanization—no one and no organization (particularly research institutions) should be engaging or tolerating the dehumanization of other individuals or groups. (And they should be holding themselves accountable for past acts of dehumanization, such as the Tuskegee Syphilis study or the Stanford prison experiment.) Lastly, it is the responsibility of all U.S. organizations and residents to defend democracy. Democracy is not synonymous with capitalism. Democracy requires impartial justice systems and equal access to opportunity, as well as social services that allow for a healthy and thriving public.

Moreover, capitalism *can* exist without democracy, as it does in China, Russia, and Turkey. If social science is on the side of democracy while being neutral about capitalism, then sound social science will be seen as science in the public interest, not "advocacy." Unfortunately, Big Business has been propping up capitalism at the expense of democracy for years now, and scientists have been willing to embrace capitalism enthusiastically through kick-backs and commissions while arguing against taking political positions. In my opinion, the marriage of science, particularly behavioral science, and capitalism is a far greater threat to our society than the one between science and advocacy.

***Minal Bopaiah** is an author, speaker and equity strategist. She is the founder of Brevity & Wit, a strategy and design firm committed to designing a more inclusive world.*

2.2: How can behavioral science inform advocacy

In my opinion, the function of science is not for advocacy. But science can lay the groundwork for the most effective forms of advocacy. The literature is full of fantastic work about collective action, persuasion, and communication. This is why I think scientists have one of the most important roles in policy decisions.

Not only can they provide the most useful insights into the actual science behind the policy, but they can provide the tools to help make that policy a success. Politics has famously been described as the "art of the possible" and science offers a roadmap for turning these possibilities into realities.

Jay Van Bavel *is an associate professor of psychology and neural science at New York University, an affiliate at the Stern School of Business in Management and Organizations, and director of the Social Identity & Morality Lab.*

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There tends to be tension between science and advocacy: Science is about facts, advocacy is about emotions, right? But scientists and advocates usually share a common goal to create positive change in the world.

Science and advocacy can actually be complementary. They make up for each other's shortcomings. Science can uncover problems and solutions, but facts alone, or even incorporated into an emotional message, are less likely to change minds or spark action than emotional appeals. On the other hand, advocates' role is often to push for change, even sometimes pushing for more than is possible, but moving the debate so that at least some positive change occurs, even if it is not all that is needed.

Ultimately I think science and advocacy need each other in order to be as effective as possible. Why? Most scientists do not want to conduct science for the sake of science alone. They want their findings to be used to improve lives. Advocates often are the drivers of change. Advocates want the world to be materially better for people, so they generally want to advocate for what is proven to work. More collaboration between scientists and advocates could be a powerful force for good.

Josh Wright *is the executive director of ideas42.*

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One of the basic principles of behavior change is to make it easy. Yet when it comes to connecting science and policy, that's often forgotten.

Many social and behavioral scientists want to influence social and political issues. Yet successful advocacy often depends on building personal relationships and being collaborative with practitioners, advocates, business leaders, elected officials, and so on outside their social networks.

In short, social and behavioral scientists and the decision-makers they want to influence often start off as strangers. And as behavioral science research shows, strangers tend to remain strangers.

Thus, in addition to thinking through normative questions about whether scientists should engage in advocacy (and what form it should take), we also need to confront the fundamental challenge of how to make these connections in the first place.

We need evidence-based methods for connecting people with diverse forms of expertise who don't already know each other, yet are working on the same social and political issues. We need to answer the question: When do strangers choose to be collaborative with each other? Then we need to take those answers and make things easier—to figure out how to design institutions that lower social barriers, thereby increasing the likelihood that behavioral scientists create rewarding new collaborative relationships with decision-makers in the first place.

***Adam Seth Levine** is the SNF Agora associate professor of Health Policy and Management at the Johns Hopkins Bloomberg School of Public Health. He is also the president and co-founder of research4impact (r4impact.org), an organization that connects researchers and practitioners.*

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If impact is the ultimate goal, then advocacy and science can't remain separate. As we set up the first behavioral insights unit within the Indian government, the question isn't whether advocacy needs to be a part of what we do, but what kind of advocacy is the most effective. In other words, what kind of communication of science leads to maximum uptake. How do we best "nudge the nudger?"

If behavior science needs to be applied, then it also needs to be marketed. And making evidence or science more attractive shouldn't be seen as affecting the work's integrity, rather it should be seen as imperative for impact. Just as a marketer communicates a product's value proposition to the user, in the same way a scientist needs to communicate an intervention's value proposition to the policymaker.

***Pooja Haldea** is a co-founder and senior advisor at the Centre for Social and Behaviour Change at Ashoka University (CSBC).*

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No doubt, many wise and accomplished people will weigh in on your central questions, which are quite challenging. So, I will artfully navigate around them and instead share two lessons from my work in behavioral science, which I believe are relevant to advocacy.

The first lesson is to focus efforts on moving people from intent to action. Advocates often spend much of their energy trying to win over those with opposing viewpoints, or preaching to the converted. Yet behavioral science suggests that the real opportunity lies with "agnostics," who have positive intentions (to vote, to donate, to volunteer, etc.), but are not consistently acting on their beliefs. We don't need to educate these people, nor persuade them to care more about a particular cause. Instead, the answer often lies in making it easier and/or providing salient and timely reminders.

A second lesson is to define issues specifically and behaviorally. Advocating for people to "act sustainably" or "become politically engaged" may sound motivating. But these goals are not actionable, until they are defined as specific and measurable behavior changes. Once this happens, we can begin to develop and test interventions and measure their success. For me, the takeaway is that advocates need to transition from speaking in inspiring generalities to facilitating specific behavioral changes.

While behavioral scientists may make different decisions regarding their personal involvement, I'd argue that the field has much to offer advocates to improve the effectiveness of their efforts.

Scott Young *is senior vice president of BVA Nudge Unit, a global behavioral science consultancy.*

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While a researcher may not choose to engage in overt advocacy or activism, we should realize that all research is inherently political. Choices are made about what research to fund and conduct, who has input into the decisions and process, and who can access the results.

I currently work at the Stockholm Environment Institute, which exists to connect science with decision-making to develop solutions for a globally sustainable future. SEI researches issues of climate, water, air and land-use, governance, economy, gender, and health. At the core of the work is stakeholder involvement and equipping partners for long-term change, and we make all of our knowledge and findings freely available to decision-makers and civil society.

I believe that denying the political nature of research, particularly by those doing it, is to live a life willfully unexamined in a way that we would not accept when conducting research. We should all be questioning our motives, our assumptions, our values and how our research fits with who we believe we are and who is and should be benefiting from our work.

Jean McKendree *is a cognitive scientist who has worked in psychology departments, research centres, and medical schools in applied education research.*

2.3: Specific ideas at the intersection of science and advocacy

Despite the important role of science in society, it is no longer sufficient for scientific papers to “speak for themselves” via traditional means of knowledge dissemination. Instead, scientists need to actively engage with political decision makers to better connect scientific research with policy development. One emerging model is Science Meets Parliament, through which social and natural scientists around the nation meet directly with Members of Parliament and Senators on the Parliament Hill to discuss each other’s work, to humanize the person behind scientific papers and policies, and to find common ground despite ideological differences.

Such interactions have mutual benefits to both scientists and policymakers, including an opportunity for the scientist to learn about how policymaking works and where science can contribute the most to policy, and an opportunity for the policymaker to learn about scientific findings firsthand, research funding landscape, and the role of science in the economy and communities. This model helps build trust and a relationship between the science and policy communities and break down disciplinary and sectoral silos. Ultimately, the Science Meets Parliament model helps foster the involvement of science in policymaking and the recognition of science as an apolitical public benefit.

Jiaying Zhao *is the Canada research chair and an associate professor in the department of psychology and the Institute for Resources, Environment and Sustainability at the University of British Columbia.*

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Knowledge is a form of power. Thus, behavioral science instructors must engage in equity-minded pedagogy. In doing so, instructors take personal responsibility for and hold their institutions accountable for student success outcomes and arm students with the skills they need to navigate and address social injustices inside and outside the classroom.

We see a great opportunity to shift research methods courses toward being more equity-minded to intervene on the academic pipeline and increase the representation of minoritized students in behavioral science. Research methods courses are often presented as apolitical and objective; however, how we design these courses signals to students what is and is not important within the field. By choosing (consciously or unconsciously) to highlight only a small segment of human experiences (i.e., White and WEIRD experiences), instructors signal to minoritized students that their experiences are outside the scope of behavioral science.

Further, many social injustices directly affect minoritized communities. By employing equity-minded pedagogy, instructors make space for students to learn about and take part in work directly affecting them. In contexts where decision-makers do not reflect the communities they serve, inequities emerge—just consider the racial inequities rampant in COVID-19 vaccine distributions. Advocating for inclusive pedagogy gives room for all students to grow as decision-makers, potentially yielding positive outcomes for historically underrepresented and underserved communities.

In short, equity-minded pedagogy is one lens through which we can advocate for our students and, in turn, arm them with the skills they need to advocate for themselves.

Tissyana Camacho *is an assistant professor of child and adolescent development at California State University, Northridge.* ***Margaret Echelbarger*** *is a postdoctoral principal researcher at the Center for Decision Research at the University of Chicago Booth School of Business.*

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Advocacy and organizing must seek to address problems identified by the people impacted by them. People are the experts of their experiences, and they have both the moral authority and the capacity to identify solutions that fit their shared values and self-interest. However, within that framework, there remains an essential role for science to play.

For the last 20 years, I supported youth of color and people from low-income communities as they fought for education equity. Their knowledge and insights were profound and always drove our campaigns. In our most successful efforts, the solutions people put forth were then assessed based on external data to ensure that people had the benefit of information beyond their own experience. For example, during a campaign to eliminate the overuse of suspensions and expulsions, a national advocacy group was consulted to learn what solutions had proven effective across the country. The result was a push for evidence-based restorative justice programs.

The students impacted by the policies in question retained the power to choose the solution for which they fought. The inclusion of science and data only strengthened their efforts. In the absence of such information, people risk pushing for policies that might be ineffective and, at worst, counter to the intended outcome.

Adam Levner *is the former executive director and cofounder of Critical Exposure, a Washington D.C. nonprofit that teaches youth how to use the power of photography and organizing to fight for educational equity.*

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Behavioral science is at a unique intersection between science and advocacy, particularly when it comes to inequality, diversity, and inclusion. With growing support for critical movements like #BlackLivesMatter and an explosion of different, but potentially problematic, labels for traditionally underrepresented groups (e.g., BIPOC, BAME, URM), scientists must engage in meaningful activism and advocate for racial terminology that is inclusive and appropriate.

Knowing what racial labels to use in different contexts will not only inform policymakers at a societal level but also companies, at an organizational level, who regularly use these labels in diversity reporting and to track progress toward diversity targets. At an individual level, it is also important to determine how these labels are perceived and how they influence intra- and inter-group outcomes.

What we have taken away from our ongoing research on this topic is that behavioral scientists are well-positioned to contribute to important advocacy and policy around racial labeling at all levels. In order to ensure a distinction between evidence-based vs. political recommendations, scientists should use nationally-representative surveys and experiments to collect data from both majority and minority group members when exploring these issues.

Racial labeling is a concern for the present; one that requires both science and advocacy. Giving back this power of self-definition to minority groups is especially critical now to combat some of the disempowerment and oppression that those groups have historically faced, and behavioral scientists can partner with these groups to co-create the research needed to give them back their voices.

Grusha Agarwal *is a Ph.D. student in the organizational behavior and human* *resource management department (OBHRM) at the University of Toronto Rotman School of Management.* ***Joyce He*** *is an incoming assistant professor of management and organizations at UCLA Anderson School of Management.* ***Sonia Kang*** *is the Canada Research Chair in identity, diversity, and inclusion, and an associate professor of organizational behavior and HR management at the University of Toronto.*

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The scientific process is framed as a neutral endeavor investigating an objective world beyond an unbiased researcher. Science, however, is located within sociopolitical contexts shaping questions, methods and study result interpretation. Science and scientists may maintain power relations by the knowledge they disseminate. For example, police killings of Black Americans are vastly underreported and repeatedly attributed to medical causes inherent to the victim, as was the case with George Floyd. With racial capitalism, White supremacy, and hetero-patriarchy responsible for the public health issues of our time, science must be at the forefront of confronting these systems.

The de-contextualization, ahistorical nature, and absence of marginalized perspectives leads to distorted knowledge production. Scientists own the means of knowledge production and research subjects are often subordinated to their agendas. An alternative paradigm for researchers, practitioners, and, most importantly, subjugated communities is to co-produce knowledge. This sits activism within the scientific method rather than merely a potential outcome emergent from its results.

To bridge society's fractures across race, gender and class, a community advisory board, supported by academics and practitioners, can define research questions and methods. Our collective—PRAXIS (Practicing Radical Action by Expanding Inclusive Societies)—defines priority areas for violence research and related political actions. The community members central to this project have experienced structural racism in the form of police violence, the carceral state. In this alternative model, community voices predominate and the means of knowledge production are transferred to communities historically serving as "subjects." For PRAXIS, science and activism do mix and are both necessary to translate research into action.

Mihir Chaudhary *is a surgical resident training to be a trauma surgeon at University of California–San Francisco East Bay.* ***Tanya Zakrison*** *is a trauma surgeon, associate professor, and director of critical trauma research at the University of Chicago.* ***Craig Hardaway*** *is a resident of Chicago, outreach worker with Chicago CRED, and part of the PRAXIS community advisory board.*

Tell us what you think. Get in touch here.

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Cintia Hinojosa

Cintia Hinojosa is a Ph.D. student of behavioral science at the University of Chicago Booth School of Business and a graduate student writing fellow at *Behavioral Scientist.*

Twitter Website

Evan Nesterak

Evan Nesterak is the co-founder and editor-in-chief of the *Behavioral Scientist*.

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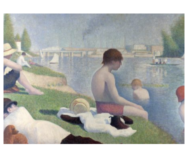

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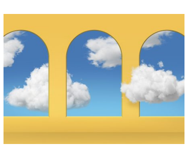

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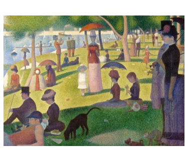

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